Deployment

Day 5 — When the model meets the clinic

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Where we are on the spine

Handoff — participants recap their Day 4 method rationale.

- **Yesterday:** a definition of success → a built model.
- **Today:** a model that works in a notebook → a system that **survives a clinic.**

Internal validity (Days 3–4) → deployment validity (today)

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The deployment gap

- **A strong notebook number is not clinical impact.**
- Most models that perform well are **never deployed** — or **fail when they are.**
- The gap between a model and a system that changes care is where this day lives.

Day 1: Pete's line, returned

Validation is not deployment

A model that validates beautifully can fail the moment it meets a workflow.

- Validation tests the **model.**
- Deployment tests the **system, the workflow, and the people.**
- The cliff between them is steep — and most projects never see it.

FIG F5.1 **The validation cliff.** A line rising smoothly through "internal validation" then dropping at the "deployment" boundary — annotated with what changes at the cliff (data pipeline, workflow, users). *See build guide F5.1.*

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REFRAMING THE OBJECT

What "deployment" actually means

The system around the model

Deployment is a **sociotechnical system**, not a model artifact.

- Data pipelines feeding the model in **real time.**
- Interfaces and alerts that carry the output to a human.
- Escalation paths for when the prediction fires.
- Governance that owns the thing once it is live.

FIG F5.2 **The model is one box.** A system diagram where the ML model is a single small box embedded in a larger loop: live data → model → interface → clinician → action → outcome → (back to data), with governance wrapping the whole. *See build guide F5.2.*

The model is **one component** — often the **easy 20%.** The system is the deliverable.

The mechanisms — *why* it breaks

Not "deployment is hard" — the precise mechanisms this audience should name:

- **Performative prediction** — a deployed risk score changes clinician behavior, which changes the outcome it predicts; the model becomes part of its own data-generating process (Perdomo et al., 2020).
- **Feedback loops** — acting on the prediction *censors* the labels you'd need to evaluate it.
- **Automation bias** — the human "in the loop" over-trusts the output and stops being an independent check.

Day 1: prediction is performative

Workflow integration

- A model that does not **fit the clinical workflow** does not get used — regardless of accuracy.
- **Where** in the workflow the prediction lands determines its value.
- Integration is **designed**, not bolted on afterward.

Worked example — right prediction, wrong moment

- A sepsis prediction that arrives **after** the clinician has already decided is **worthless** — even if correct.
- The right output at the wrong moment is **noise** that erodes trust.
- Timing is a first-class design variable, not an implementation detail.

Clinician adoption and trust

- Adoption is a **human problem** — alert fatigue, override behavior, eroding trust.
- **A correct prediction that is ignored has zero clinical value.**
- Trust is earned slowly and lost fast — one bad alert can sink a tool.

↩ Day 3: alarm fatigue from class imbalance

Worked example — accurate, and still ignored

The most instructive failure: nothing was *wrong* with the model.

- Metrics held. Calibration held. The audit passed.
- And clinicians **stopped looking** — wrong interface, wrong cadence, no trust built.
- A model with great numbers and zero adoption has **zero clinical value.**

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THE MOST VALUABLE SLIDE

A model that validated — then failed on the ward

The validation-to-ward gap — a worked case

- A model that **validated well** — strong metrics, clean study — then **failed in practice.**
- The **canonical public case:** the widely-deployed Epic Sepsis Model, externally validated, showed poor discrimination and **alert fatigue** at the bedside (Wong et al., 2021).
- The **counter-example: CHARTWatch** cut unanticipated mortality ~26% — but only after **9 months of silent testing** and serious change management (Verma et al., *CMAJ* 2024).
- Your own story makes it visceral; the public case makes it **generalizable.**

FIG F5.3 **Your case, one image.** A before/after or timeline of the specific deployment you narrate — what the metrics said vs. what happened on the ward. Build from your own program's data. *See build guide F5.3.*

Transport fails — the same model, a new hospital

- A sepsis model can perform very differently **across hospitals** (Lyons et al., 2023).
- Case mix, documentation habits, and pipelines differ — the model meets a **different world.**
- "It worked at our site" is a hypothesis about your site, not a property of the model.

FIG F5.4 **Performance across sites.** A dot/forest plot: the same model's metric at site A (where it was built) vs. several deployment sites, with visible spread and degradation. *See build guide F5.4.*

Day 2: distribution shift · Day 3: external validity

KEEPING A DEPLOYED MODEL ALIVE

Monitoring, drift, and audit

Monitoring — when you can't see the labels

Models decay. The hard part isn't that — it's that **you usually can't measure it live:**

- Outcome labels are **absent or delayed** (and feedback loops censor them), so you often **can't compute accuracy** in real time.
- You monitor **proxies** — input drift, prediction drift, calibration on the slow trickle of labels — and accept they're imperfect.
- Live performance is partly **unverifiable** — design for that, don't pretend otherwise.

FIG F5.5 **Drift over time.** A monitoring chart: a tracked signal (input/prediction drift, or calibration) trending past a control band toward a "retrain / investigate" trigger. *See build guide F5.5.*

Day 3: evaluating without ground truth

Auditing during deployment

- **Pre-deployment evaluation ≠ live accountability.** Both are required.
- Ongoing audit, incident reporting, a clear owner when something goes wrong.
- **Report** development/evaluation transparently — **TRIPOD+AI** (2024) is a *reporting* guideline, not a live-deployment standard.
- **Govern the lifecycle** with fit-for-purpose frameworks — **DECIDE-AI**, **CONSORT-AI / SPIRIT-AI**, **FUTURE-AI**, **NIST AI RMF**, **FDA GMLP** — and map it (the **algorithm journey map**, Boag et al., 2024).

Equity — "fair" is a choice, not a checkbox

- A deployed model can **systematically disadvantage** a group while looking fine in aggregate (Obermeyer et al., 2019; Pfohl et al., 2024 for LLMs).
- The hard truth this audience needs: **calibration and error-rate balance are mathematically incompatible** except in degenerate cases (Kleinberg et al. 2016; Chouldechova 2017) — you *choose* among fairness criteria, you don't satisfy all.
- And subgroup monitoring inherits Day 3's **power and multiplicity** problems.

↩ Day 2: who is missing · Day 3: disaggregation & multiplicity

Who you need to deploy at all

The deployment team

You do not deploy alone. Who you need:

- **Clinical champions** — credibility and workflow insight.
- **Informatics & IT** — pipelines, integration, uptime.
- **Governance** — risk, compliance, sign-off.
- **Leadership** — budget and the mandate to change practice.

Deployment is **staffed**, not solo.

→ Day 6: collaborators & resourcing

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BEFORE IT GOES LIVE

Stress-test your own project

Red-teaming a model's implications

Anticipate failure modes **before** deployment:

- **Subgroup harms** — who is worse off if it works as designed?
- **Gaming** — how is the output exploited once people know it exists?
- **Workflow mismatch** — where does it land at the wrong moment?
- **Complementarity is not free** — human + AI often fails to beat the better of the two alone; design for it explicitly (Yu et al., 2024).

> Ask: who is harmed if this **works**? Who is harmed if it **fails**?

Case studies — successes and failures

- Two or three **concrete deployments** from a live program.
- The **unglamorous determinants** — workflow, timing, staffing, trust.
- The pattern across them: deployment is decided by the **boring details.**

Frontier — can you ever know it helped?

- "Outcomes improved after we deployed it" is **confounded** — co-interventions, secular trends, and the feedback loop all move with it.
- Knowing it *helped* is a **causal** claim → it needs an RCT of the ML intervention, or a credible quasi-experiment (Plana et al., 2022).
- Most deployed models are **never** evaluated this way. Decide up front whether yours will be.

↩ Day 3: evidentiary standards · Day 1: prediction vs. causation

Red-team your own project

Today's deliverable — Deployment plan + risk assessment.
The **system** around your model, the **workflow** it lands in, your **monitoring/audit** plan, and an honest **risk assessment** — subgroup harms, failure modes, who is accountable.

Next: 11:00 small group — your peers red-team your project · 1:30 guest (Deb Raji) — auditing & accountability · 2:45 fireside — the clinician & patient voice.